《2021－2022 秋学期信号与系统Ⅰ》

一、填空题

1、$y(k)=f(k)\cos\pi k$，该系统是（线性、非线性）（时变，时不变）（稳定/不稳定）的系统

2、$f(k)=\cos[2k]$ 则该信号的周期为 $\underline{\infty}$ 　　$w=2$　$T=\frac{2\pi}{2}=\pi$，非周期

3、对信号压缩了后增强了（高频，低频）分量

4、$f(5-2t)=2\delta(t-3)$ 则 $\int_{-\infty}^{+\infty} f(t)(t+3)dt=\underline{8}$

$$f(t)=2\delta(\frac{5}{2}-\frac{t}{2}+3)=2\delta(-\frac{1}{2}-\frac{t}{2})=4\delta(t+1)$$

$$\int_{-\infty}^{+\infty} f(t)(t+3)dt=\int_{-\infty}^{+\infty} 4(t+3)\delta(t+1)dt=4(t+3)\Big|_{t=-1}=8$$

5、LTI 系统 $h(t)=e^{-5t}U(t)$，在激励 $3\delta(t)+2\delta(t-2)$ 下的零状态响应为：$\underline{3e^{-5t}U(t)+2e^{-5(t-2)}U(t-2)}$

① $h(t)*f(t) \longleftrightarrow H(jw)\cdot F(jw)$

$\Rightarrow$ ② $U(t)*\delta(t)=U(t)$

6、$2G_{10}(k)$ 门信号的能量是 $\underline{40J}$

$$E=\sum_{k=-5}^{5}|2G_{10}(k)|^2=4\times 10=40J$$

7、$H(j\omega)=G_{2\omega_c}(\omega)$低通滤波器对$f(t)=5\cos5t+10\cos100t$无失真传输，则截止频率为 100Hz

8、$F(j\omega)=2U(1-\omega)$对应$f(t)=$ $\delta(t)+\frac{e^{jt}}{j\pi t}$

$U(t)\longleftrightarrow \pi\delta(\omega)+\frac{1}{j\omega}$ $\pi\delta(t)+\frac{1}{jt}\longleftrightarrow 2\pi U(-\omega)$

$2U(-\omega)\longleftrightarrow \delta(t)+\frac{1}{j\pi t}$ $2U(1-\omega)\longleftrightarrow e^{jt}(\delta(t)+\frac{1}{j\pi t})$

9、$y(k)+3y(k-1)+2y(k-2)=2^kU(k)$

$y(0)=0$，$y(1)=2$，求初始状态 $y(-1)=0$ $y(-2)=0.5$

10、$f(t)=5[\cos(100\pi t)]^2$的周期，频谱不混叠的最小频率 200Hz

$f(t)=\frac{5}{2}+\frac{5}{2}\cos(200\pi t)$ $\omega_m=200\pi$ $f_m=100$ $f_H=200$ (奈奎斯特)

11、$H(s)=\frac{e^{-s}}{s^2+(k-5)s+1}$，当$k=$ 5 时临界稳定

此时单位冲激响应为 $\sin(t-1)U(t-1)$

$s^2+1=0$ $s=j\Rightarrow$ 临界稳定

$\sin\omega_0 t\longleftrightarrow\frac{\omega_0}{s^2+\omega_0^2}$ $f(t-t_0)\longleftrightarrow F(s)e^{-t_0 s}$

二、某系统框图如图所示，$h(k)=\delta(k)+5\delta(k-1)$

$h_2(k)=U(k)-U(k-2)$，求 $h_1(k)$

$$f(t) \longrightarrow \boxed{h_1(t)} \longrightarrow \boxed{h_2(t)} \longrightarrow y(t)$$

$$H(z)=Z[h(k)]=1+5z^{-1}$$

$$H_2(z)=Z[h_2(k)]=1+z^{-1}$$

$$\begin{cases} H(z)=H_1(z)\cdot H_2(z) \\ h(k)=h_1(k)*h_2(k) \end{cases}$$

$$1+5z^{-1}=(1+z^{-1})H_1(z) \Rightarrow H_1(z)=\frac{1+\frac{5}{z}}{1+\frac{1}{z}}=1+\frac{4}{z+1}$$

$$h_1(k)=Z^{-1}\left[1+\frac{4}{z+1}\right]=Z^{-1}\left[z^{-1}\left(z+\frac{4z}{z+1}\right)\right]$$

$$=\delta(k-1)*\left[\delta(k+1)+4(-1)^k U(k)\right]$$

$$=\delta(k)+4(-1)^{k-1}U(k-1)$$

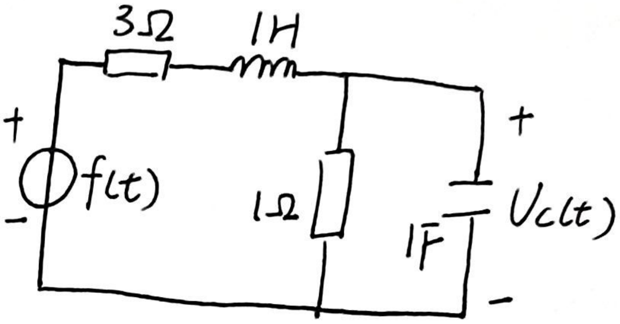
三、如图所示零状态电路，$U_c(t)$为响应，激励

$f(t)=12U(t)\,V$，求

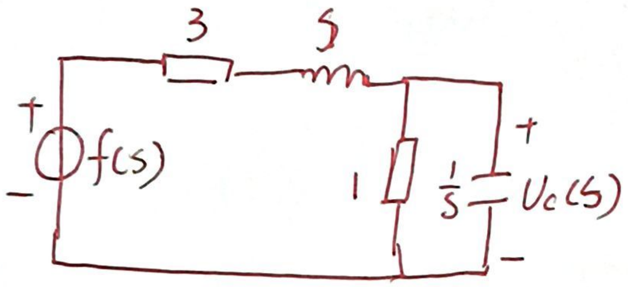
(1) $H(s)$及电路自然频率

(2) 关于$U_c(t)$的微分方程

(3) 零状态响应$U_c(t)$



(1) ⇒



⇒ $$U_c(s)=\frac{\frac{1}{s+1}}{3+s+\frac{1}{s+1}}F(s)$$

$$=\frac{1}{s^2+4s+4}F(s)$$

$$H(s)=\frac{1}{s^2+4s+4}$$

自然频率为-2

(2) 由 $(s^2+4s+4)U_c(s)=F(s)$

$\Rightarrow U_c''(t)+4U_c'(t)+4U_c(t)=f(t)$

(3) $$U_{fc}(s)=H(s)F(s)=\frac{12}{s}\frac{1}{(s+2)^2}=\frac{3}{s}-\frac{3}{s+2}-\frac{6}{(s+2)^2}$$

$$F(s)=\frac{12}{s}$$

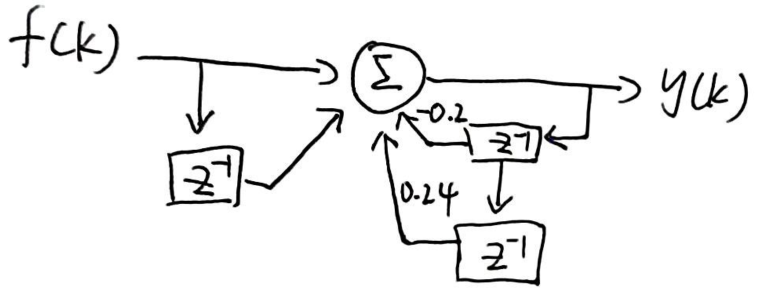
$$U_{cf}(t)=\mathcal{L}^{-1}[U_{fc}(t)]=(3-3e^{-2}-6te^{-2})U(t))$$

四 如图所示为离散因果系统的系统框图.由系统框图求

(1) $H(z)$并判别系统稳定性

(2) 单位序列响应$h(k)$

(3) 输入$f(k)=2+3\cos(\frac{\pi}{2}k+45^\circ)$时的稳态响应



(1) $f(k)+f(k-1)-0.2y(k-1)+0.24y(k-2)=y(k)$

$z_1=0.4$

$z_2=-0.6$

$|z|<1$

∴稳定

$y(k)+0.2y(k-1)-0.24y(k-2)=f(k)+f(k-1)$

$(1+0.2z^{-1}-0.24z^{-2})Y(z)=(1+z^{-1})F(z)$

$$H(z)=\frac{1+z^{-1}}{1+0.2z^{-1}-0.24z^{-2}}=\frac{z^2+z}{z^2+0.2z-0.24}=\frac{1.4z}{z-0.4}-\frac{0.4z}{z+0.6}$$

(2) $h(k)=\mathcal{Z}^{-1}[H(z)]=1.4(0.4)^k u(k)-0.4(-0.6)^k u(k)$

(3) $H(e^{j\omega})=H(z)\big|_{z=e^{j\omega}}=\dfrac{e^{2j\omega}+e^{j\omega}}{e^{2j\omega}+0.2e^{j\omega}-0.24}$

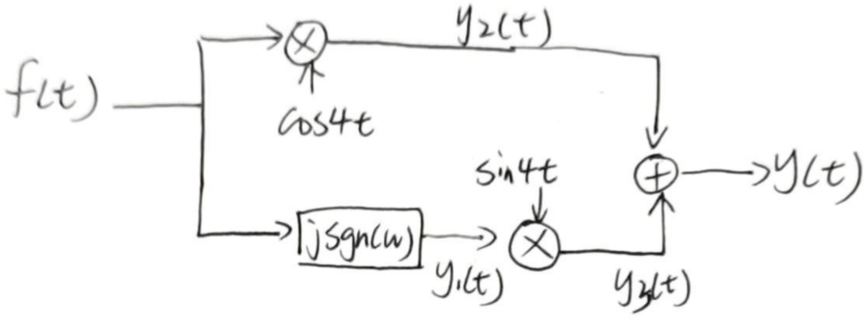
① 当$\omega=0$时，$|H(e^{j\omega})|=\dfrac{1+1}{1+0.2-0.24}=2.1$ $\varphi(0)=0$

② 当$\omega=\frac{\pi}{2}$时，$|H(e^{j\omega})|=\left|\dfrac{-1+j}{-1.24+j0.2}\right|=1.13$ $\varphi(\frac{\pi}{2})=-35.8^\circ$

$=4.2+3.39\cos(\frac{\pi}{2}k+9.2^\circ)$

$\therefore y(k)=2\times2.1+3\times1.13\cos(\frac{\pi}{2}k+45^\circ-35.8^\circ)$

五 $F(j\omega)=G_4(\omega)$ 求 $Y_2(j\omega)$ $Y_3(j\omega)$ $Y(j\omega)$ 和它们频谱图



$$Y_1(j\omega)=F(j\omega)\cdot[j\,\mathrm{sgn}(\omega)]=j[G_2(\omega-1)-G_2(\omega+1)]$$

$$Y_2(j\omega)=F(j\omega)*\pi[\delta(\omega-4)+\delta(\omega+4)]=\pi[G_4(\omega-4)+G_4(\omega+4)]$$

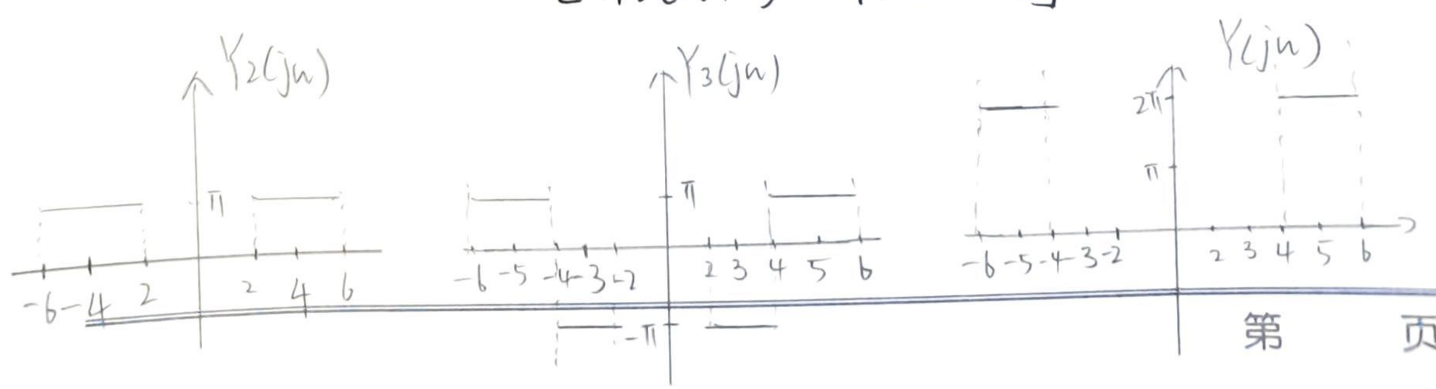
$$Y_3(j\omega)=Y_1(j\omega)*j\pi[\delta(\omega+4)-\delta(\omega-4)]$$

$$=-\pi[G_2(\omega+3)-G_2(\omega+5)-G_2(\omega-5)+G_2(\omega-3)]$$

$$=-\pi\{[G_2(\omega+3)+G_2(\omega-3)]-[G_2(\omega+5)+G_2(\omega-5)]\}$$

$$Y(j\omega)=Y_2(j\omega)+Y_3(j\omega)$$

$$=\pi\{-[G_2(\omega+3)+G_2(\omega-3)]+[G_4(\omega-4)+G_4(\omega+4)]-[G_2(\omega+5)+G_2(\omega-5)]\}$$

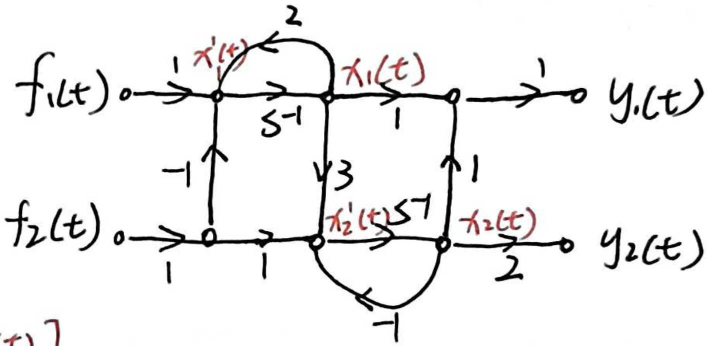


六 已知系统的信号流图如图所示

(1) 求系统矩阵形式的状态方程和输出方程

(2) 求系统自然频率，判断系统稳定性

(3) 求系统状态转移矩阵 e^{At}



(1)

$$\begin{cases} x_1'(t)=2x_1(t)+f_1(t)-f_2(t) \\ x_2'(t)=3x_1(t)-x_2(t)+f_2(t) \end{cases}$$

$$\begin{bmatrix} x_1'(t) \\ x_2'(t) \end{bmatrix}=\begin{bmatrix} 2 & 0 \\ 3 & -1 \end{bmatrix}\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}+\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} f_1(t) \\ f_2(t) \end{bmatrix}$$

$$\begin{cases} y_1(t)=x_1(t)+x_2(t) \\ y_2(t)=x_2(t) \end{cases}$$

$$\begin{bmatrix} y_1(t) \\ y_2(t) \end{bmatrix}=\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$$

$\Longrightarrow A\begin{bmatrix} 2 & 0 \\ 3 & -1 \end{bmatrix}$

$B\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$

$C\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

$D\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

(2)

$$|SI-A|=\begin{vmatrix} S-2 & 0 \\ -3 & S+1 \end{vmatrix}=(S-2)(S+1)$$

解：$\Longrightarrow P_1=-1 \quad P_2=2$

有自然频率为 $P_1=-1$，$P_2=2$.

P_1 位于左半平面但 P_2 位于右半平面

∴系统不稳定

13) $\Phi(s) = (sI - A)^{-1} = \frac{1}{(s-2)(s+1)} \begin{bmatrix} s+1 & 0 \\ 3 & s-2 \end{bmatrix}$

$$= \begin{bmatrix} \frac{1}{s-2} & 0 \\ \frac{3}{(s-2)(s+1)} & \frac{1}{s+1} \end{bmatrix} = \begin{bmatrix} \frac{1}{s-2} & 0 \\ \frac{-1}{s+1} + \frac{1}{s+2} & \frac{1}{s+1} \end{bmatrix}$$

转移矩阵 $e^{At} = \mathcal{L}^{-1}[\Phi(s)] = \begin{bmatrix} e^{2t} & 0 \\ e^{2t} - e^{-t} & e^{-t} \end{bmatrix} U(t)$